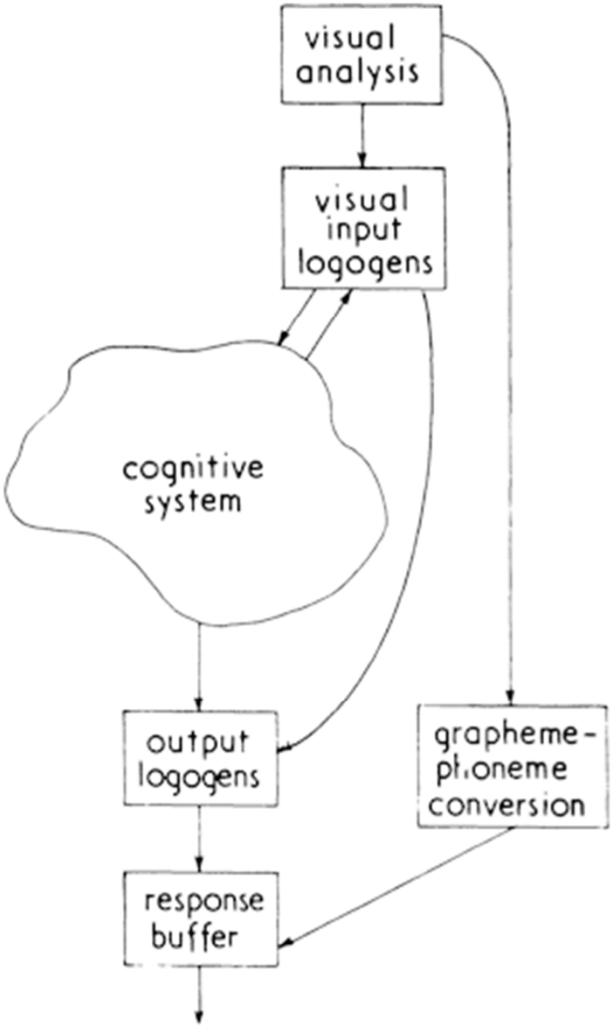
Reading 2: Computational models

Dual route models of visual word recognition

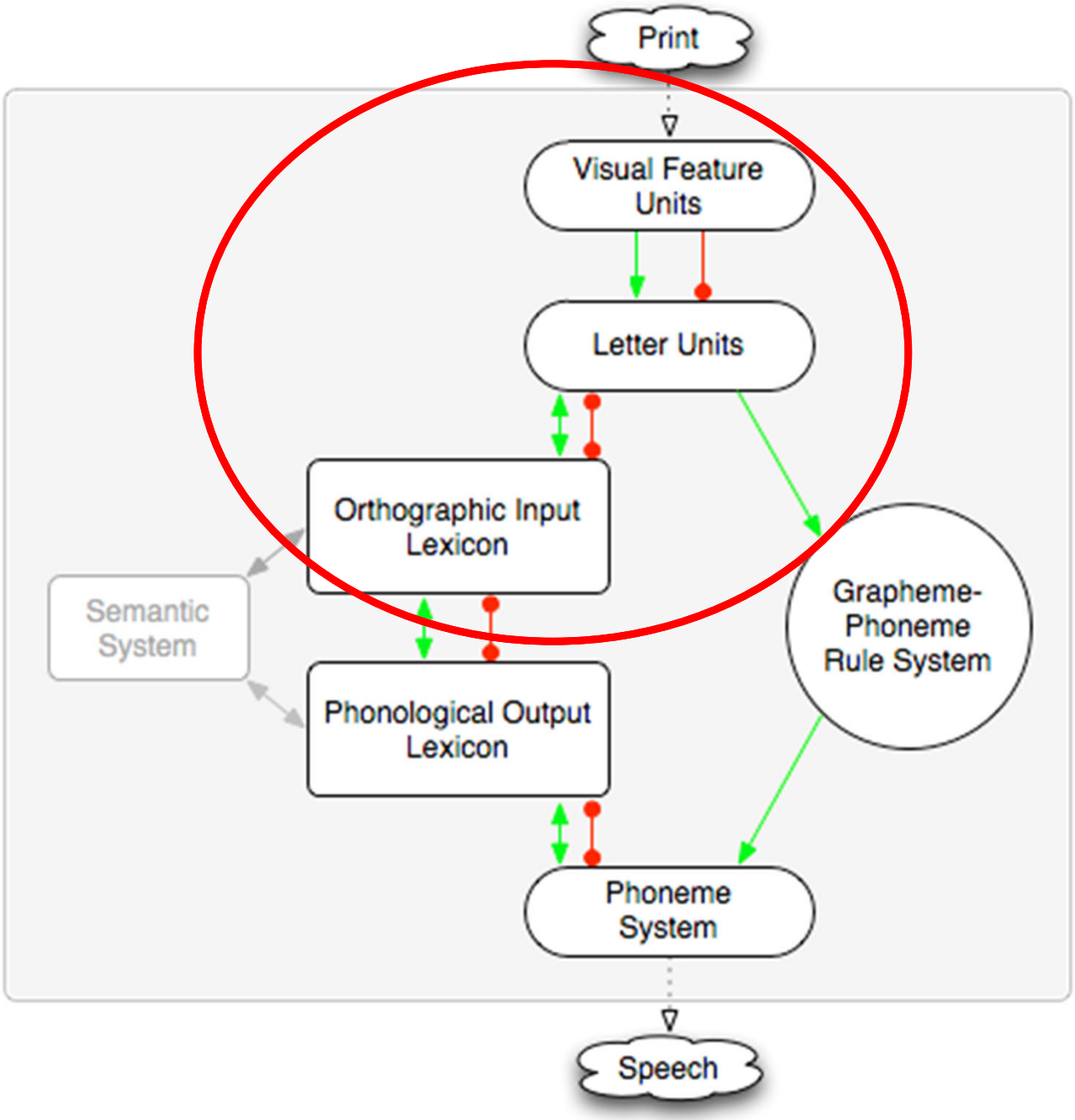


This general form of model is motivated by the phenomena we reviewed in the last lecture:

- pseudoword reading is fast and automatic
- word reading is faster
- reading is fastest for high-frequency words
- accuracy and response time are also impacted by "regularity"
- frequency and regularity interact
- pseudoword reading and irregular word reading are dissociable in acquired dyslexia

Thus, the model hypothesize

- a *sublexical route* for grapheme-to-phoneme conversion (decoding)
- a *lexical route* for recognizing real words



This is the "DRC -- Dual Route Cascaded" model.

Some key points:

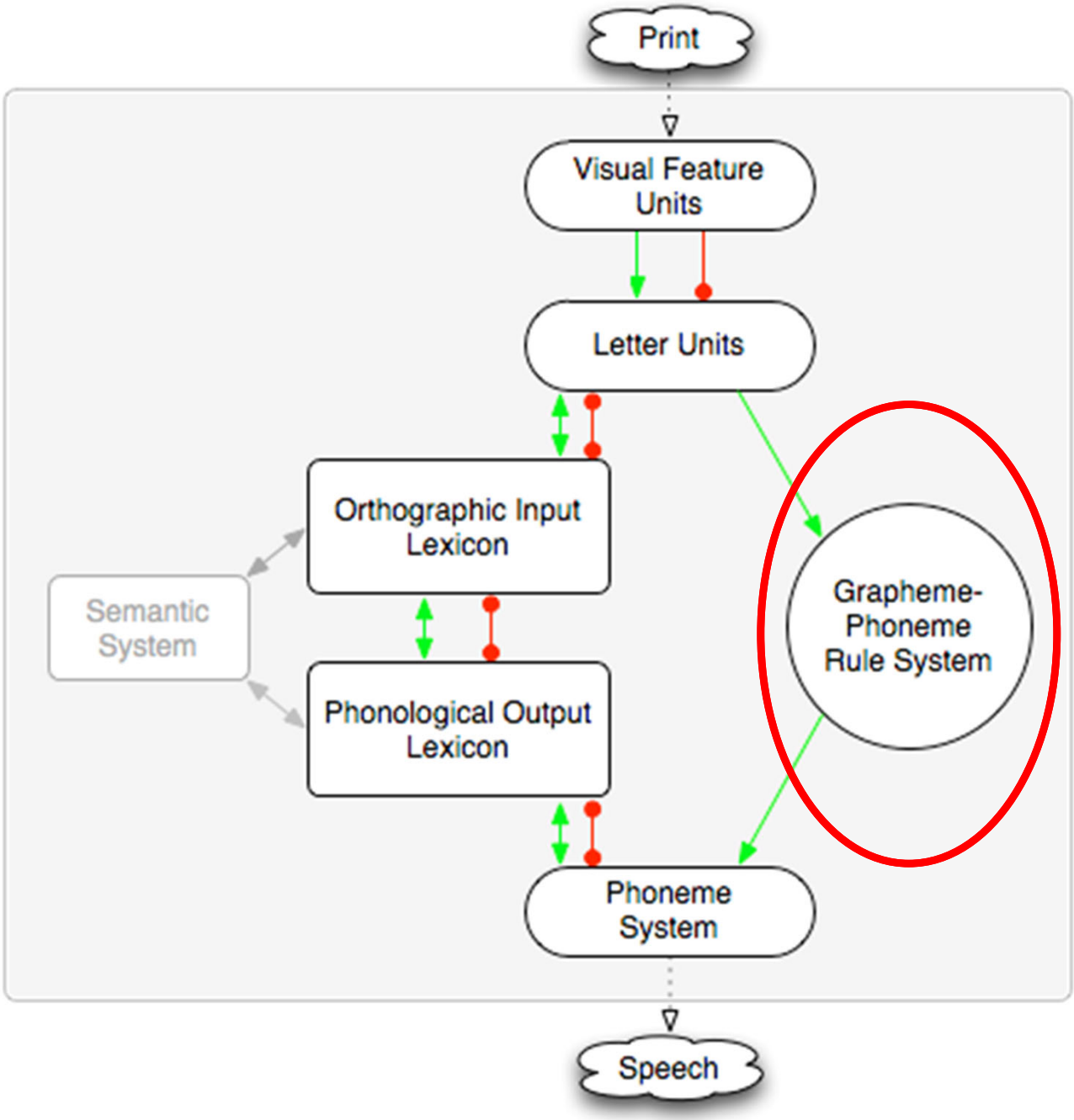
The portion circled in red is *just* the Interactive Activation model.

There is no implementation of semantics, since both the orthographic and phonological lexicons are independent of semantics.

The Grapheme-Phoneme Rule System has completely different processing assumptions from the interactive portion of the model.

(image credit: http://www.maccs.mq.edu.au/~ssaunder/DRC/)

Coltheart et al. (2001, Psych Review)



Some important features of the Grapheme-Phoneme Rule System:

Its rules are activated serially, letter by letter.

It is not affected by frequency of occurrence in any way.

It has no feedback.

It does not interact with the lexical route, except in the phoneme response buffer, where it is connected to the same output units (and therefore can compete to activate different patterns).

(image credit: http://www.maccs.mq.edu.au/~ssaunder/DRC/)

Coltheart et al. (2001, Psych Review)

What are the GPC rules like?

Appendix B: DRC's Nonlexical Rules

Grapheme	Phoneme	Position A=all b=beginning m=middle e=end
Multiletter Rules		
arre	#	A
augh	9	A
eare	7	A
eere	7	A
eigh	1	A
ough	9	A
tsch	J	e
urre	3	A
ai.e	1	A
ar.e	#	A
aw.e	9	A
ea.e	i	A
ee.e	i	A
er.e	3	A
ie.e	i	A
<illegible>	ς	^
ll	l	m,e
mb	m	m,e
mm	m	m,e
mn	m	m,e
ng	N	m,e
nn	n	m,e
oa	5	A
oe	5	A
oh	5	A
oi	4	A
oo	u	A
or	9	A
ou	6	A
ow	6	A
oy	4	A
ph	f	A
pp	p	m,e
ps	s	b
re	r	m,e
rh	r	b

Context-Sensitive Rules: The grapheme outside the square brackets in Column 1 is converted to the phoneme in Column 2 in the presence of the preceding or following context specified in the square brackets. [V]=any vowel; [C]=any consonant.

gu[V]	g	b
n[k]	N	A
[q]u	w	m
g[e]	_	b
c[e]	s	A
c[i]	s	A

(you also need some meta-rules for when to apply rules at different levels of analysis, e.g., larger-unit rules supersede smaller-unit rules)

Actually, the rules aren't totally independent of statistics.

They're often based on the most frequent pronunciation for a letter or combination of letters.

But they are *fixed* and *categorical.* So, even though the rule for -OUGH has weaker statistical support than the rule for -EIGH, they are both rules and behave the same way during processing.

	EIGH	{EIGH: (final != T)}	OUGH	{OUGH: (final != T)}
/eɪ/	0.667	1	-	-
/aɪ/	0.333	-	-	-
/ɔ/	-	-	0.53	0.27
/əf/	-	-	0.16	0.27
/oʊ/	-	-	0.16	0.18
/aʊ/	-	-	0.11	0.18
/u/	-	-	0.05	0.09

These rules are kind of silly

…by which I mean that they seem to miss a lot of cases.

This was seen as a *feature*, though, not a *bug.*

We could certainly come up with *better rules* if we took statistics into account more flexibly. (In fact the DRC's rules went through several revisions to improve its nonword reading.)

But the argument from the DRC is that people don't use such finely-tuned rules for the GPC route.

If they did, you wouldn't see the pattern of regularization errors that you see in surface dyslexia, or the slow-down in performance that you see for low-frequency exception words.

These rules are kind of silly

Remember when we talked about models of syntax?

We saw that parsing strategies like “late closure” and “minimal attachment” were motivated by consideration of syntactic complexity *only*.

The idea was to explain syntactic processing as *informationally encapsulated* from world knowledge, lexical co-occurrence statistics, etc.

The dual-route models are designed to be similarly *modular.*

The GPC chugs along and produces correct pronunciations for many words, incorrect responses to some “exceptions” to its rules. Occasionally, we can observe its operation when:

-people produce “regularization” errors

- we force people to name pseudowords

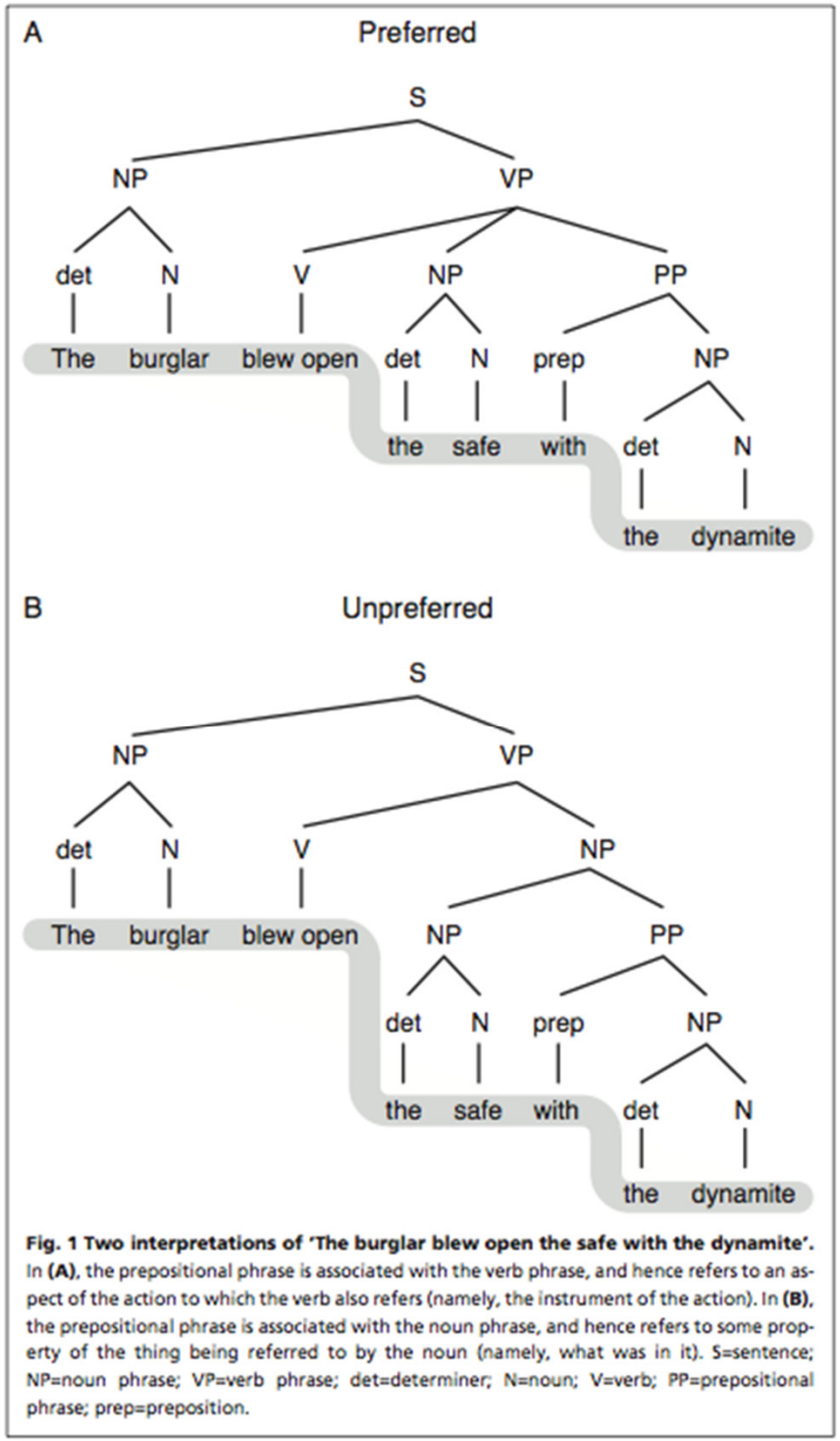


Fig. 1 Two interpretations of ‘The burglar blew open the safe with the dynamite’. In **(A)**, the prepositional phrase is associated with the verb phrase, and hence refers to an aspect of the action to which the verb also refers (namely, the instrument of the action). In **(B)**, the prepositional phrase is associated with the noun phrase, and hence refers to some property of the thing being referred to by the noun (namely, what was in it). S=sentence; NP=noun phrase; VP=verb phrase; det=determiner; N=noun; V=verb; PP=prepositional phrase; prep=preposition.

The DRC can simulate a lot of phenomena in visual word recognition.

prefer a different approach. Essentially, our view is that the past quarter of a century of empirical and theoretical research on reading has provided us with good reasons for proposing a particular architecture for the human reading system, and our preference is to rely on this body of literature, rather than on backpropagation, for ideas about what this architecture might be. We are thus adopting the same view as Grainger and Jacobs (1998) who wrote, "We therefore argue that in developing algorithmic models of cognitive phenomena, the major source of constraint is currently provided by human behavioral data" (p. 24).

Reading Aloud

1. Frequency effect
2. Lexicality effect
3. Regularity effect
4. Interaction of regularity with frequency
5. Interaction of regularity with position of irregularity
6. Consistency effect
7. Pseudohomophony effect
8. Base word frequency effect on pseudohomophone reading
9. Absence of *N* effect on pseudohomophone reading
10. Presence of *N* effect on nonword reading
11. Whammy effect
12. Strategy effects
13. Homophone and pseudohomophone priming
14. Repetition priming
15. Onset effect in masked form priming
16. Triple interaction between regularity, frequency, and repetition
17. Length effect
18. Interaction between lexicality and letter length

Lexical Decision

1. Word frequency effect
2. Pseudohomophone effect
3. Interaction between pseudohomophone effect and orthographic similarity
4. *N* effect on NO responding
5. Interaction between *N* and frequency on YES responding

Acquired Dyslexias

1. Quantitative simulation of exception word, regular word, and nonword reading in surface dyslexia, and of the effect of frequency on surface dyslexic reading of exception words.
2. Quantitative simulation of the interaction between pseudohomophony and orthographic similarity in phonological dyslexia.

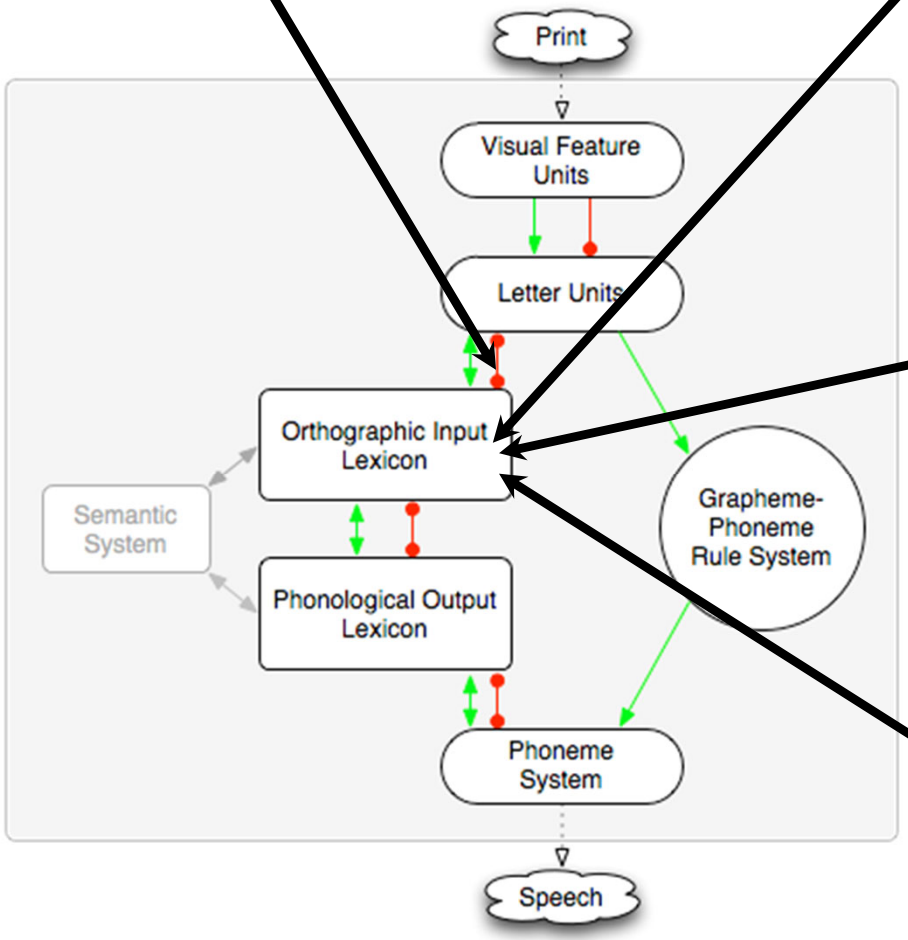
Other Effects

1. Stroop effect and its interaction with position of overlap
2. Prediction of regular-word reading accuracy from accuracy exception-word and nonword reading in normal, dyslexic, and brain-damaged children

Coltheart et al., 2001, Psych Review

This sometimes means going to extremes to make contact with data

Reduce letter-word inhibition.
(This change makes the task harder, but without it, the results make no sense.)



If any single word has an activity above a threshold of 0.69, answer “yes.”

If the activity over the orthographic input lexicon exceeds threshold S, answer “yes.”
(where S = 10 from 0-20 cycles, and is then reduced to 1.98 if activity in the orthographic lexicon exceeds 0.2 at 20 cycles).

If neither of these criteria is met by D cycles, answer “no.”
(where D = 42 unless activity in the lexicon exceeds 0.112 after 20 cycles, in which case it is set to 48. Note that this nearly twice as fast as the model normally reads aloud).

How the lexical decision task is modeled in the DRC…

Key feature of the DRC: Modularity

The GPC route and lexical route:

- each have their own processing assumptions

- are *informationally encapsulated* from one another

- are domain-specific, i.e., the GPC rules can only take letters as inputs and pass phoneme activations to the phoneme system

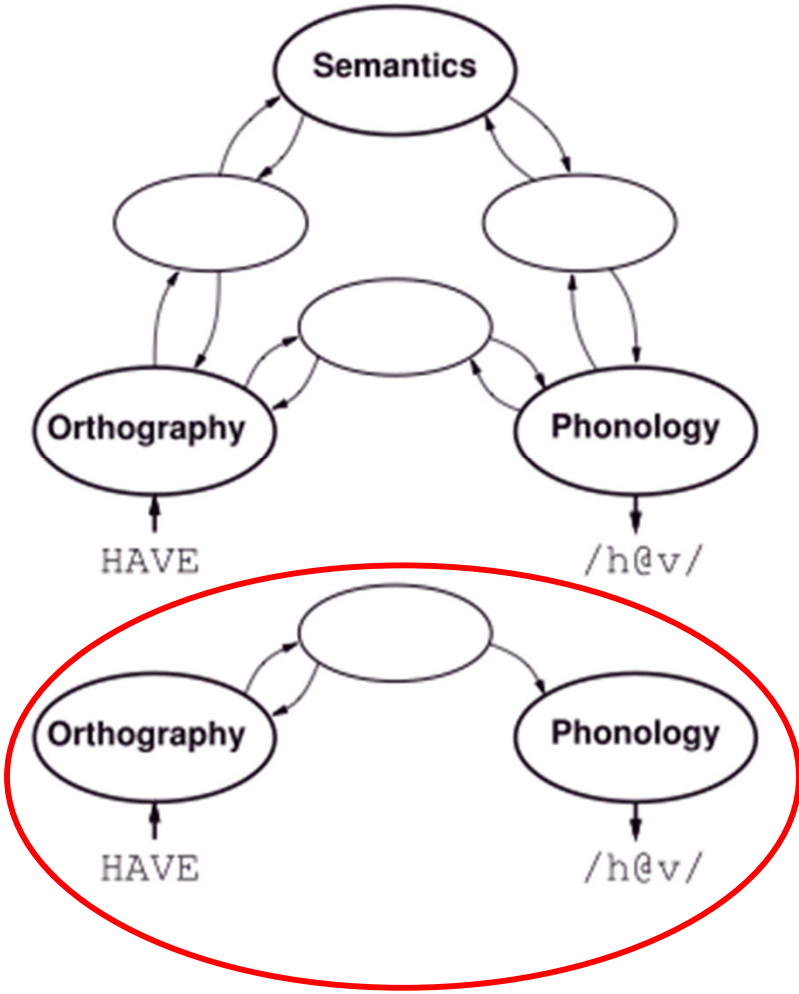
An emergentist approach: PDP models of visual word recognition

Seidenberg & McClelland (1989)

Conceived of reading skill as emerging from mappings among spelling, pronunciation, and meaning.

Modeled the *learning* of these mappings by training a simple associative network.

Studied mappings from *orthography* to *phonology* (O → P) in a single "route."



PDP models of visual word recognition

Seidenberg & McClelland (1989)

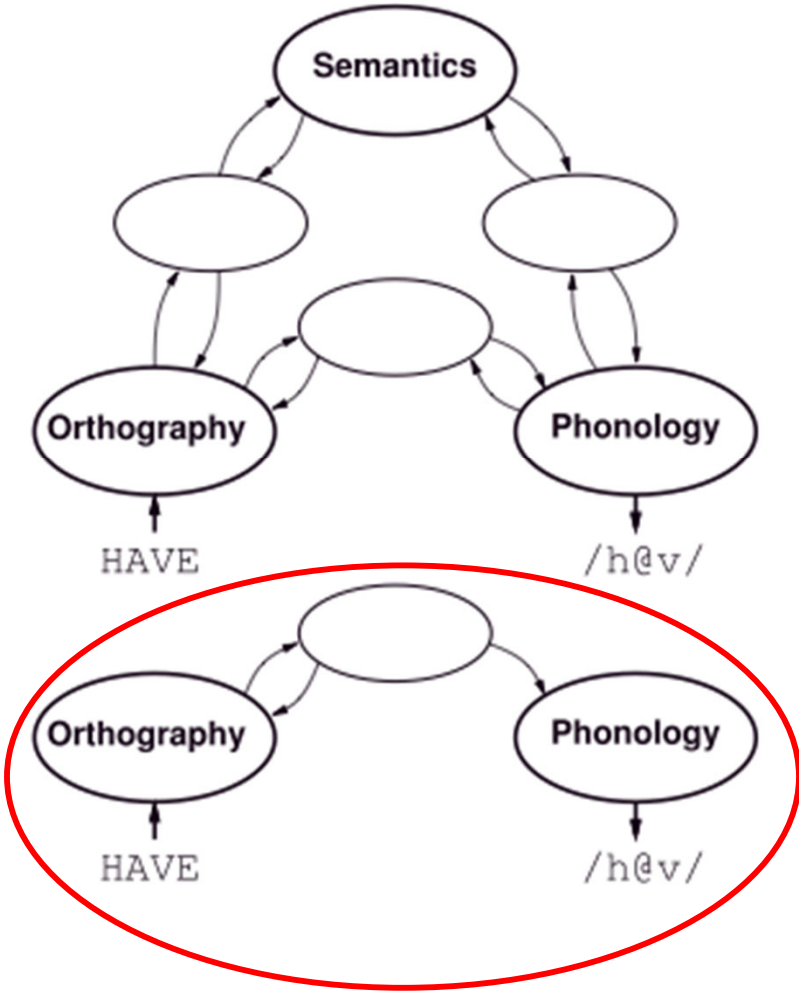
Different assumptions from dual-route models:

- "rules" are just generalizations from lots of words, not a separate process

- the "lexicon" is also distributed. Some whole-word information is stored in O → P mappings, some in other parts of the system.

Different research strategy from dual-route models:

Explain reading in terms of domain-general *learning* mechanisms, rather than reverse-engineer the system from adult/neuropsych data.

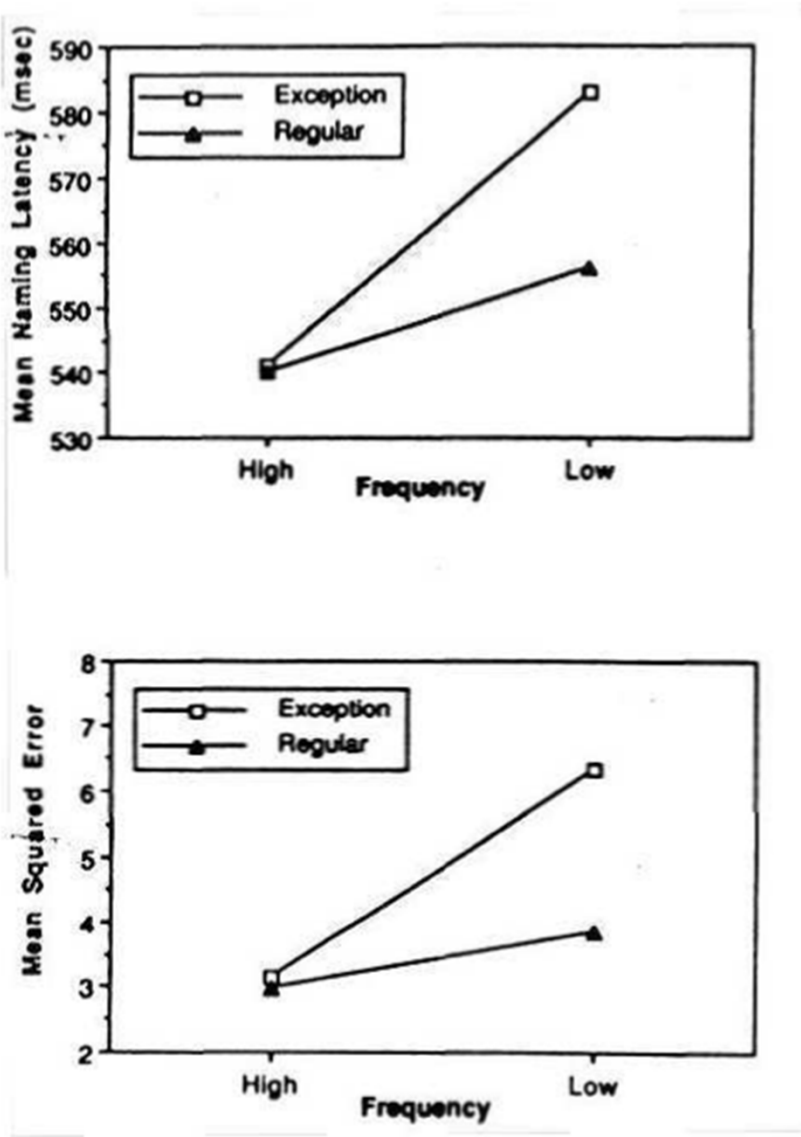


Frequency by regularity interaction

The model captured the interaction between frequency and regularity without assuming that these effects resulted from different processes.

How?

- "regular" words are *consistent* with statistical patterns from many other words.
- "exception" words are *inconsistent* with these statistics.
- frequency also impacts how well words are learned. So, highly frequent words are less effected by inconsistency with other words.



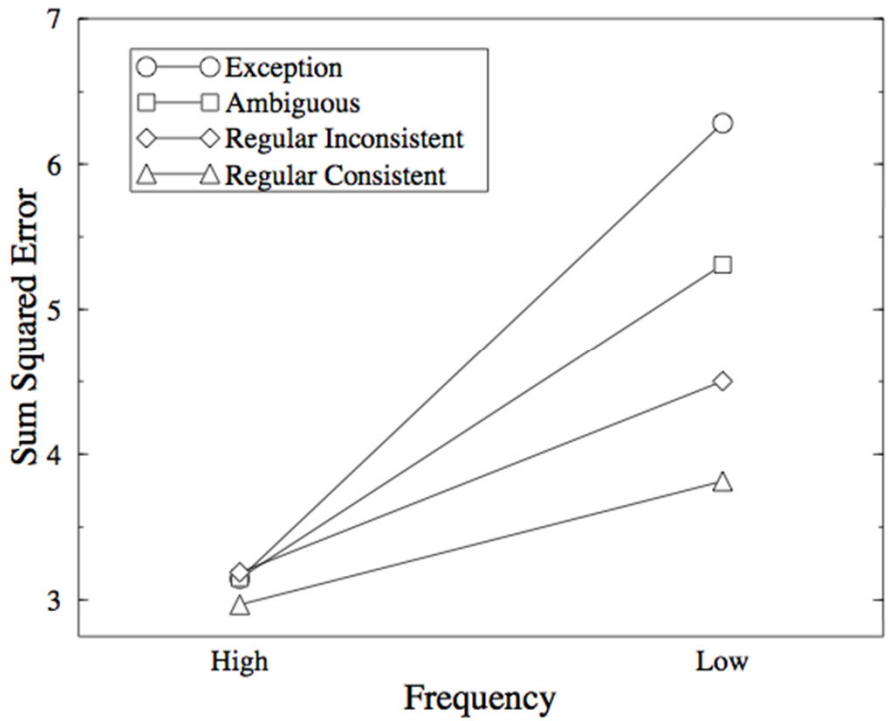
Redefining “regularity”

Regular Consistent: Many *friends* and no *enemies* [LATE, DUST]

Regular Inconsistent: Many friends but at least one enemy [GAVE (cf.HAVE), MINT (cf.PINT)]

Ambiguous: About equal numbers of friends and enemies [DOWN, KNOWN; POUR, SOUR]

Exception: Few if any friends and many enemies [HAVE (cf. GAVE), PINT (cf. MINT)]



Reading words and nonwords in the same "route"

In principle:

- nonwords named via the same set of weights used for words.

- generalization takes advantage of the same statistical patterns that give rise to regularity effects.

In practice, though, not so great…

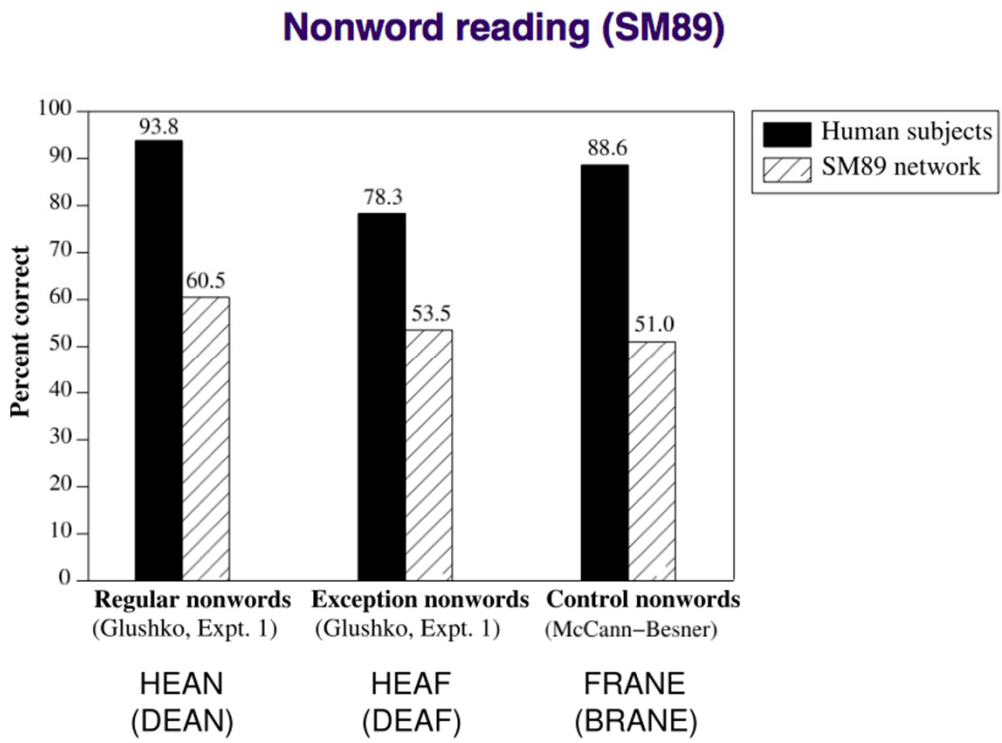
Nonword accuracy was not very good.

Why?

This is a classic case of ambiguity in model evaluation.

Was the theoretical claim -- that nonwords and exceptions could be read by the same, associative mechanism – wrong?

Or, was the problem something specific to the way the model was made?

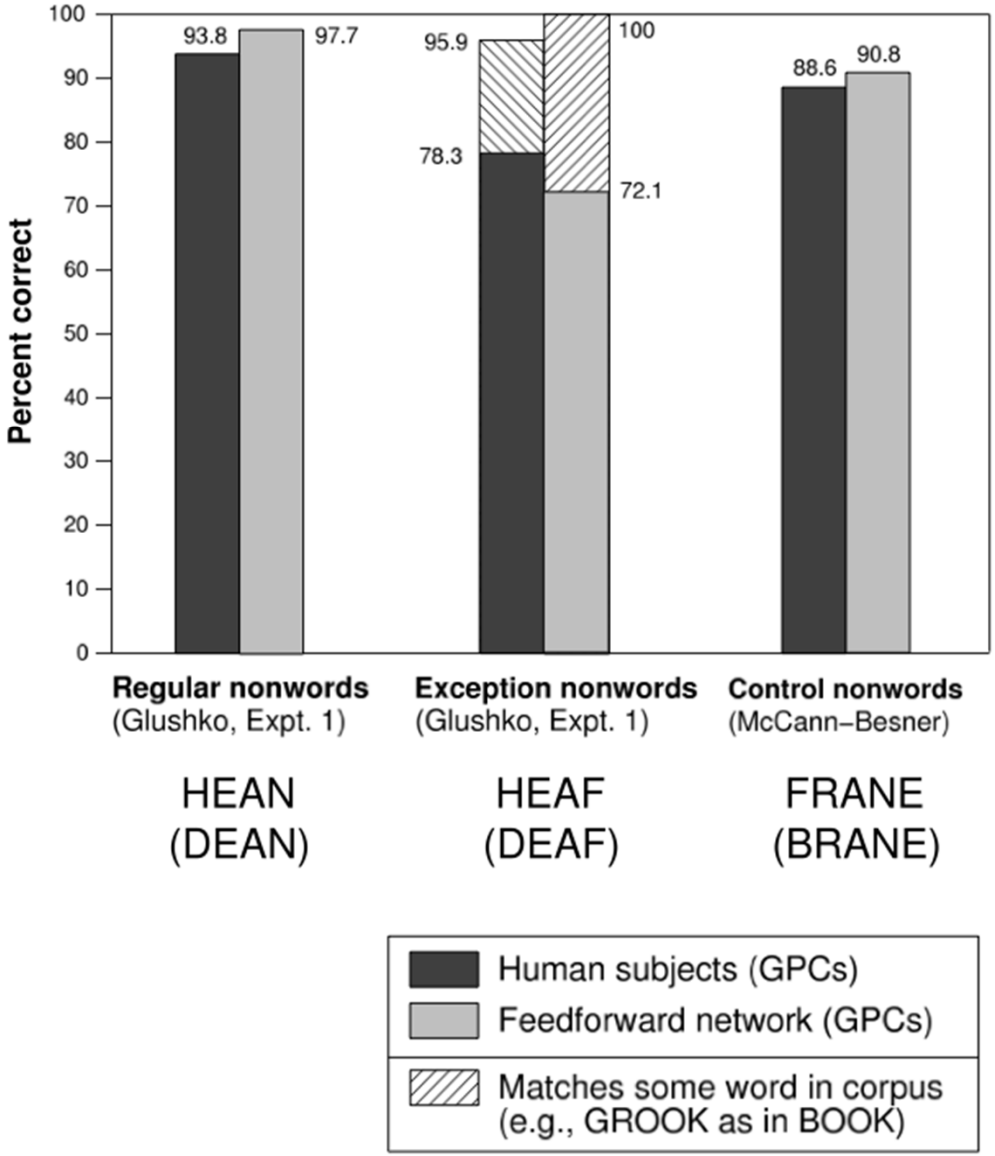


Nonword reading related to phonological representations

Many studies demonstrate a relationship between "decoding" (i.e., nonword reading) and phonological processing.

The phonological representations used in Seidenberg & McClelland were sort of weird: Wickelfeatures.

Indeed, a model by Plaut et al. (1996, Psych Review) improved the phonological representations, and produced better performance on nonwords.



How are phonological representations and nonword generalization related?

Generalization depends on similarity. (In fact this principle is inherited from behaviorism.)

Poor phonological representations (like Wickelfeatures) fail to adequately capture the similarity space of words in the language.

Word-specific information is more robust to poor representations, because it doesn’t depend entirely on generalization.

How are phonological representations and nonword generalization related?

Harm & Seidenberg (1999) investigated this more systematically, by manipulating the amount of "noise" in the phonological representations.

They were able to replicate a number of key findings in the literature on developmental dyslexia:

- relationship between *degree* of phonological deficit and *phonological dyslexia*

- relative sparing of word recognition

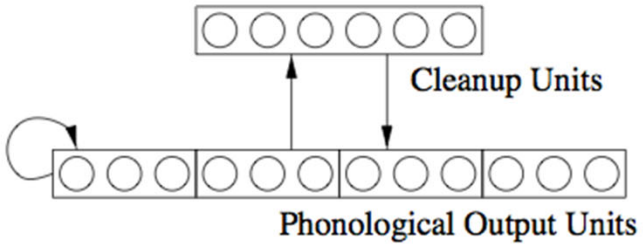


Figure 1. Architecture of the phonological attractor network.

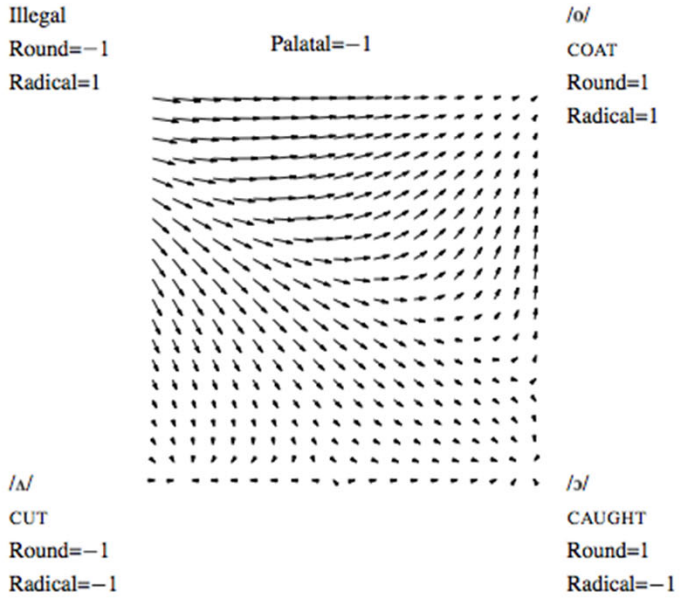


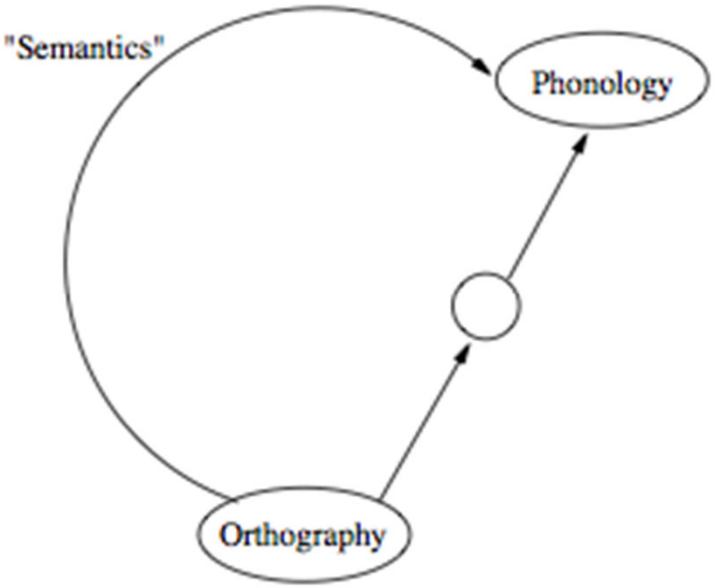
Figure 6. Phonological attractor, depicting 3 legal phonemes derived from combinations of two features: round and radical.

What about all of those double-dissociations?

We've already seen how nonword reading can be specifically impaired, despite the model's lack of a specific "nonword reading mechanism."

Plaut et al. (1996) explored the other end of this dissociation: *surface dyslexia.*

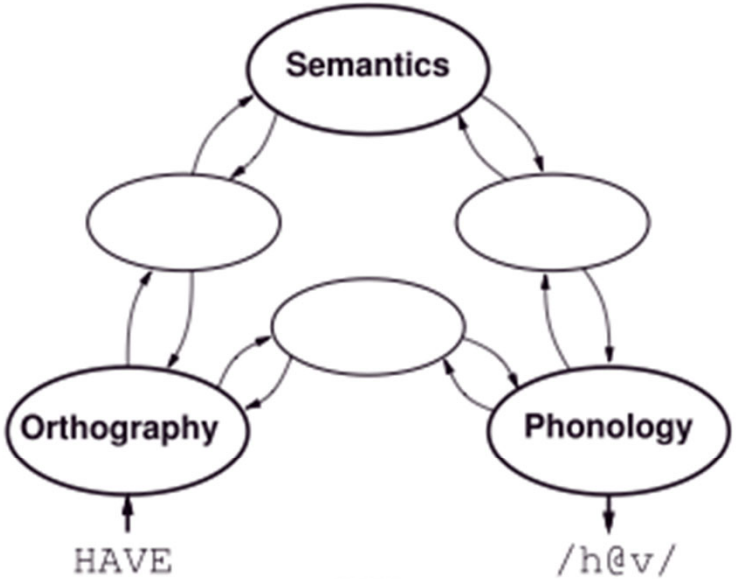
To do this, they implemented a simplified semantic system, and demonstrated that lesions to semantics resulted in a specific impairment on "exception" words.



(c) Plaut et al. 1996

Why is this not just the dual route model now?

Semantic representations are directly involved, as opposed to a theoretical commitment *to a lexicon independent of semantics.*



[A later model (Harm & Seidenberg, 2004) would implement featural semantics, and simulate the mapping of spelling to meaning.]

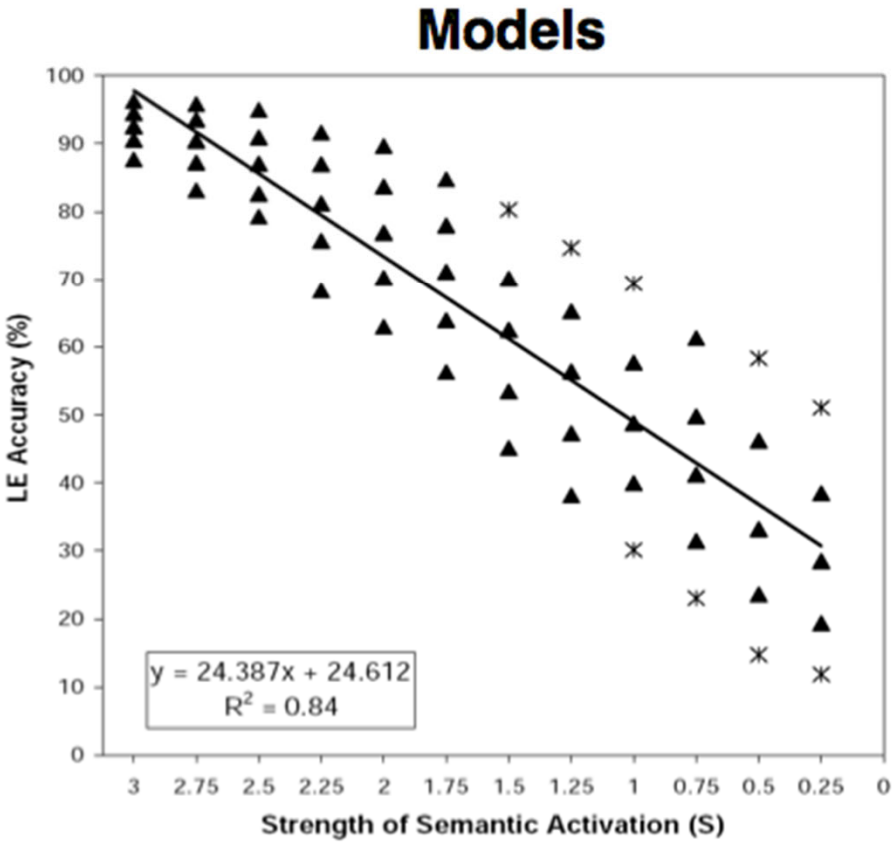
The O→P and S→P mappings have the same processing assumptions and learning algorithm. Both are probabilistic.

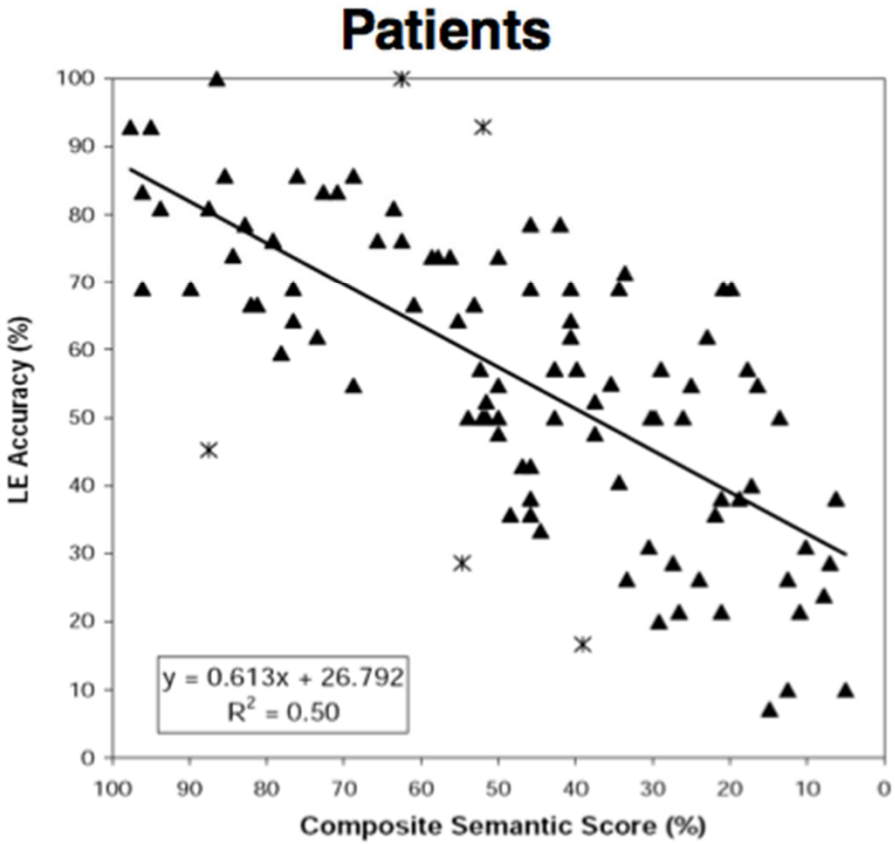
A key point of these simulations is that *these pathways interact over the course of learning*. So, the O→P pathway *develops differently in the context of an* O→S→P *pathway than it would otherwise.*

This interactivity and gradedness permits the models to capture the fact that double dissociations are never entirely "clean."

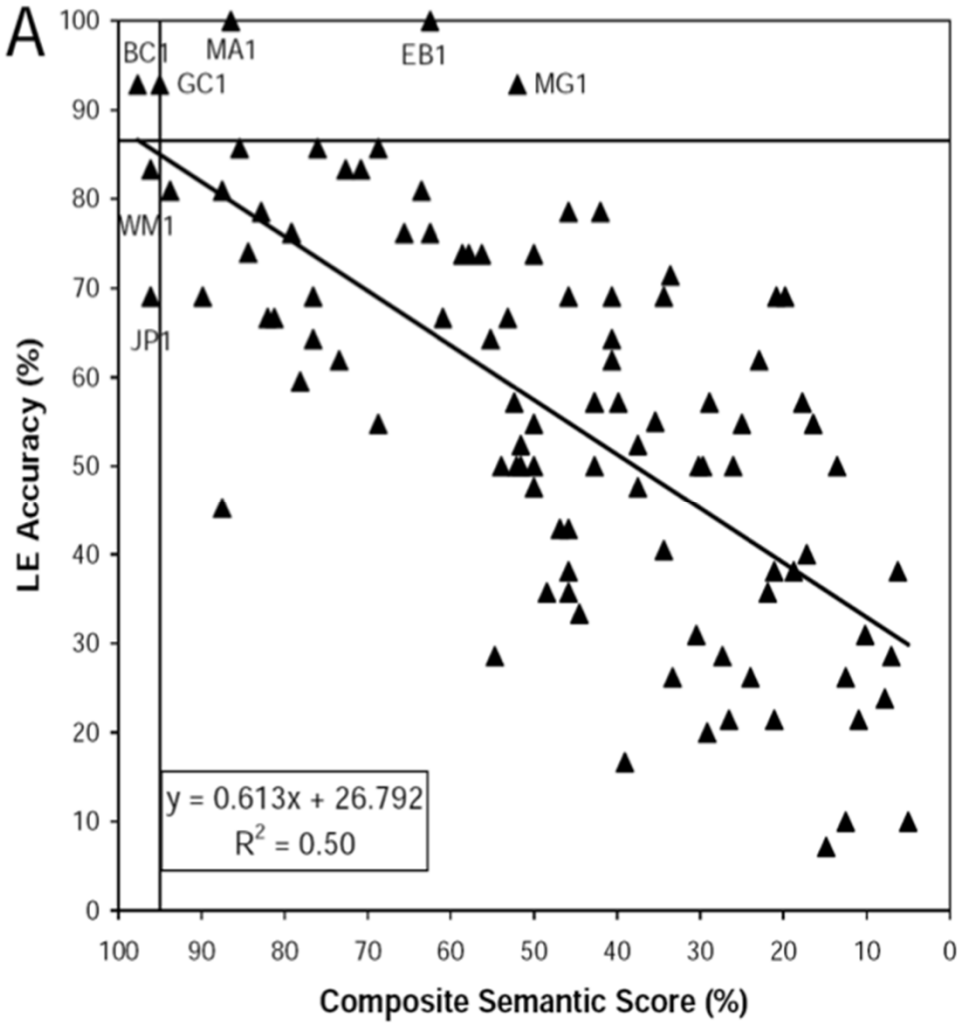
- Phonological dyslexics are not perfect on exception words.
- Surface dyslexics are not perfect on nonwords.

Model predicts a strong association between semantic abilities and exception word reading (Woollams et al., 2007)





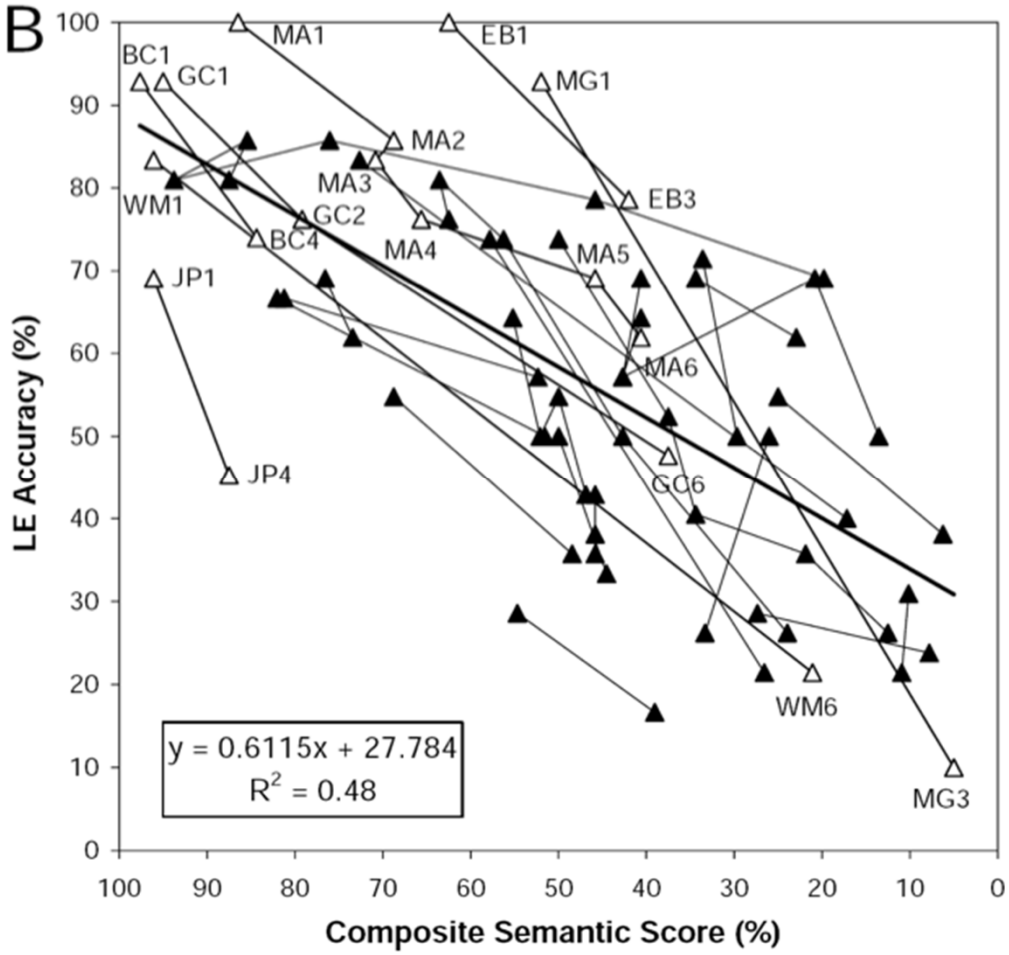
…this means the model misses a few data points



What should we make of this?

Patients to the left of the vertical line, and above the horizontal line are statistical outliers.

Woollams et al. would argue that these observations should be treated the way we normally treat outliers when evaluating the fit of a model to data…

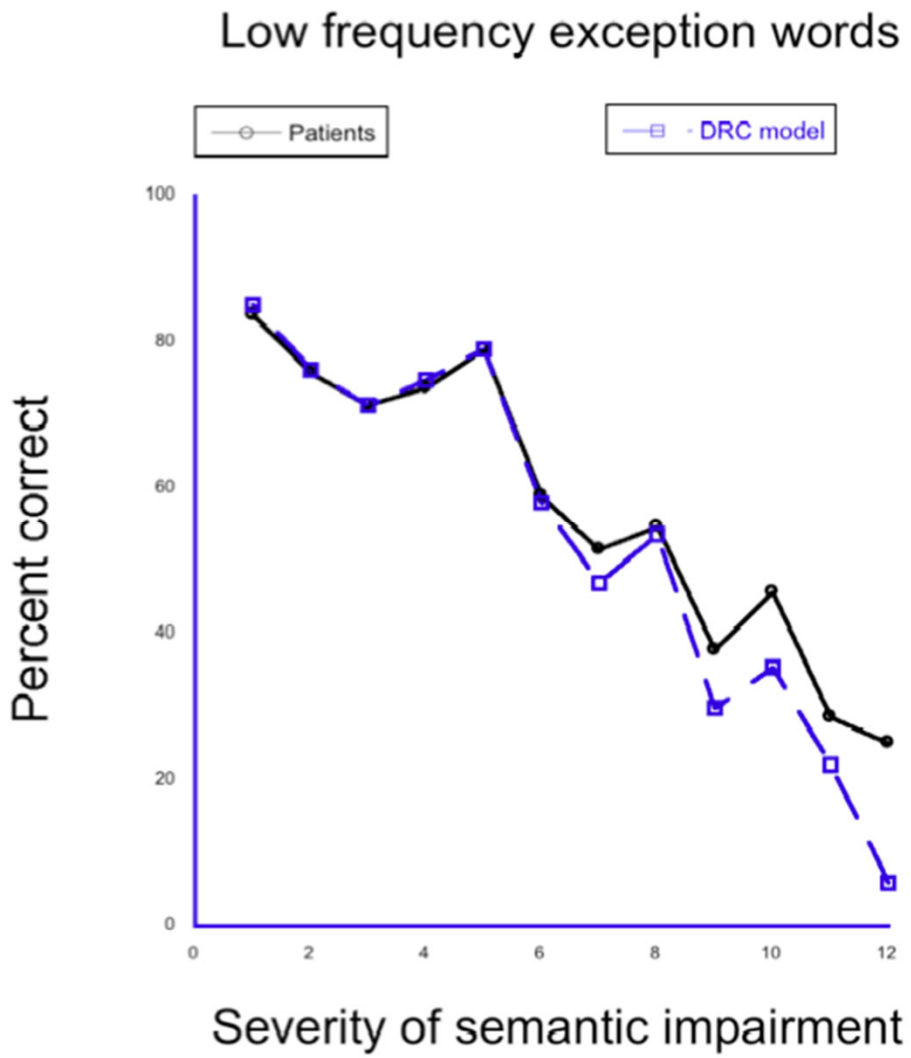


Further, because they were studying patients with a *progressive* condition (semantic dementia) they could observe how exception word reading changed as semantic abilities declined.

In most cases, when a patient's semantic abilities declined over time, her reading ability also declined.

Note also the *regression to the mean* with multiple observations. If the dissociation between semantics and exception word reading were due to idiosyncratic outliers, it would show just this pattern.

DRC simulation of the same data



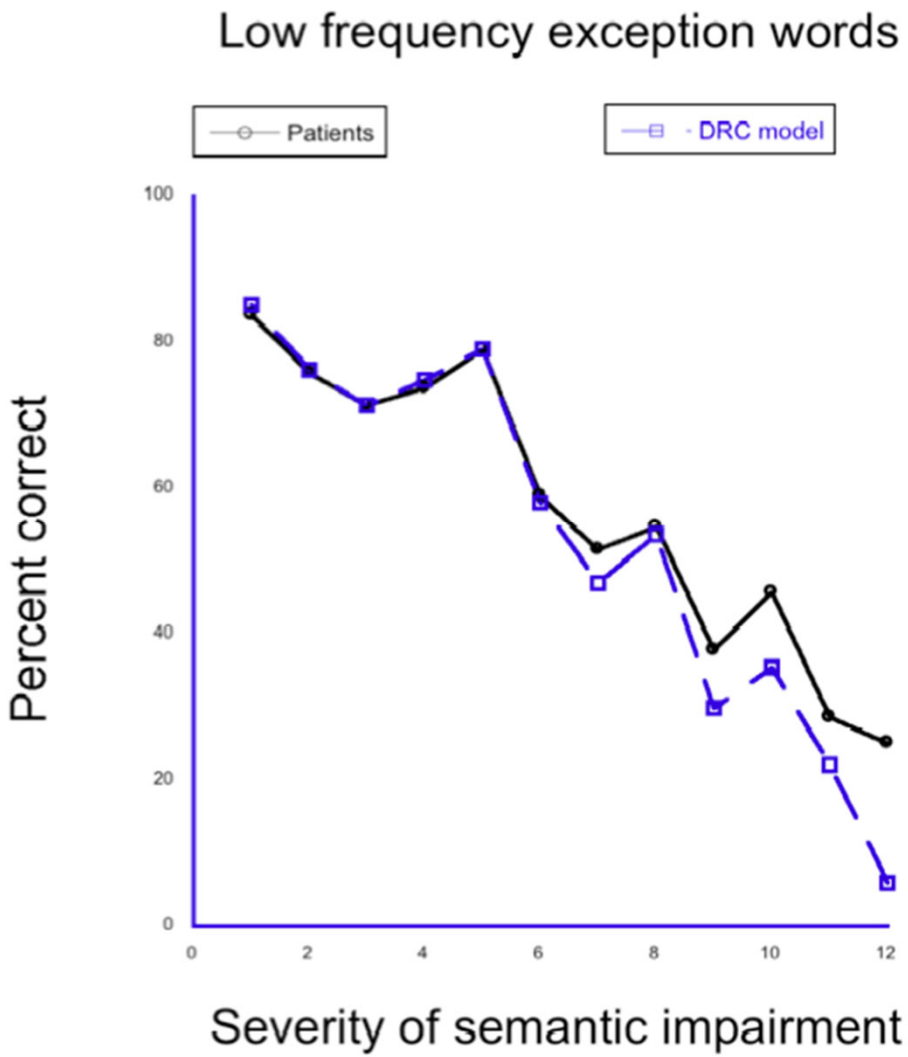
Coltheart et al. (2009, Psych Review) simulated surface dyslexia by generating a population of 40,000 model "patients" varying on two parameters:

- integrity of lexical route
- integrity of GPC rules

For each patient in the Woollams et al. data, they then selected the run of the model that best fit that patient's data.

In this way, they produce an excellent fit to the data.

Why would anyone argue that a better fit to the data is actually a *bad thing* in this case?



Comparison of dual-route and “triangle” models.

Both models cover a range of data about reading, including some basic facts:

- decoding is fast and automatic (e.g., pseudoword reading)

- words that diverge from typical spelling-to-sound mappings are harder to read

- this “regularity” or “consistency” effect interacts with word frequency

- pseudoword reading and “exception” or “inconsistent” word reading are dissociable in neuropsych patients

Comparison of dual-route and "triangle" models.

Phenomenon	Dual-Route	Triangle
Pseudoword reading and regularity effects	Dedicated "grapheme-phoneme conversion" (GPC) rule system	Statistical regularities learned over many words
Frequency effects	Distinct lexicons for input and output, separate from GPC rules	More exposure to a word = better learning for that word
Phonological dyslexia	Deficit/damage in the GPC route	Deficit/damage in phonological processing, generally
Surface dyslexia	Deficit/damage in the lexical route	Deficit/damage in semantic processes, generally

Comparison of dual-route and "triangle" models.

Phenomenon	Dual-Route	Triangle
Phonological dyslexia	Deficit/damage in the GPC route	Deficit/damage in phonological processing, generally
Surface dyslexia	Deficit/damage in the lexical route	Deficit/damage in semantic processes, generally

This is a key contrast:

Dual route models (generally) are motivated by *double-dissociations:* Pseudowords and exception words can be impaired independently, so they should be modeled as independent processes.

Models informed by statistical learning instead reveal *associations:* Phonology is important for learning/using O→P generalizations; semantics is important for words where O→P generalization is ambiguous.

For next time:

Read Joanisse (2004) “Specific Language Impairments in Children”

http://publish.uwo.ca/~marcj/joanisse-cdps-2004.pdf